Mr Dimelow

Information handling, modelling and forward planning

Topic	What students need to learn:
11 Information handling, modelling and forward planning	11.4 - Implication to designers, manufacturers and consumers of the following standards when developing designs and manufacturing products: *a) British Standards (BSI and kite mark)* *b) European (CEN and CE)* *c) International Standards (ISO)*

Standards when developing designs and manufacturing products

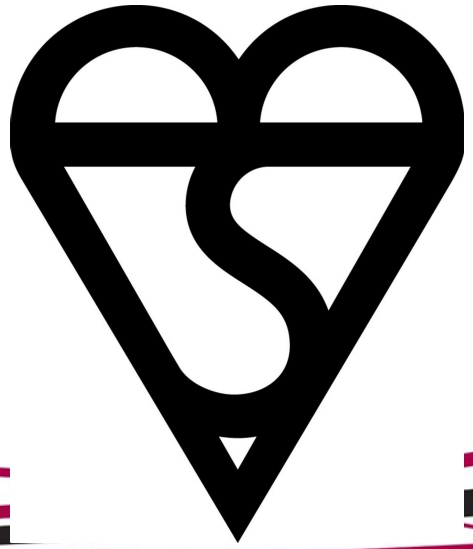
- Conforming to regularly updated national, European and international standards contributes to the commercial success of products and improves safety.
- Safety standards are compulsory for sales in specific regions, and ISO 9001 (quality management) conformity is often essential when working with other similarly certified companies all around the world.

https://technologystudent.com/prddes1/kite1.html

a) British Standards (BSI and kite mark)

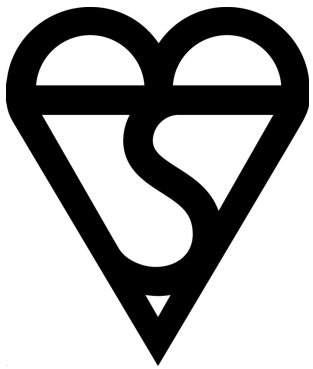
British Standards (BSI and kite mark)

- The ***British Standards Institute (BSI)*** is an example of a National Standards Body (NSB) that devises agreed standard procedures.
- Its portfolio of specifications, methods, guides, vocabularies and codes of practice currently exceeds 30,000 standards.
- The ***BSI verification Kitemark*** shows the standard has been met and is influential when consumers decide on product purchases.



British Standards (BSI and kite mark)

- The Kitemark can only be awarded if a company satisfactorily demonstrates that their manufacturing systems, tests, records and rectification procedures have been conducted correctly.



- **British Standards** accepted by the ***European Committee for Standardisation*** carry the prefix ***'BS EN'***.
- Examples of current British Standards:
  - *BS EN 71-1:2014 Safety of toys. Mechanical and physical properties.*
  - *BS EN 50361:2001 Electromagnetic effect of mobile phones*
  - *BS 1363 Three pin plugs and sockets.*

Activity - Exam Tip

- Try to find out some details of appropriate standards that apply to products you have made in school – such as lighting, toys or furniture.

- This should help you to remember them as examples.

b) European (CEN and CE)

European (CEN and CE)

- The ***European Committee for Standardisation (CEN)*** is responsible for European Standards.
- The members of the committee are the ***National Standardization Body (NSB)*** in the European Union.
- The CE mark means a product conforms to all relevant European safety standards and is mandatory for EU sales.
- The letters ***"CE"*** are the abbreviation of French phrase ***"Conformité Européene"*** which literally means ***"European Conformity".***



c) International Standards (ISO)

International Standards (ISO)

- The ***International Standards Organisation (ISO)*** consists of 150 NSB's, including BSI.
- Implements internationally recognised standards, facilitating access to foreign markets.
- The best-known ***quality management systems standard*** is ***BS EN ISO 9001***.
- Other international standards include:
  - *BS EN ISO 14000 Environmental management*
  - *BS EN ISO 50001 Energy management*

Other Standards Labels

- Various labelling systems have been adopted for products such as:

- *Lion Mark for toys*



- *Fire labels for furniture and furnishings*





- *Food ingredient labels*



Other Standards Labels

- These labels show conformity with existing standards and additional industry specific codes of practice.

Typical Mistake: making vague points about the impact of standards on safety is unlikely to gain many marks.

It would be much better to concentrate on an example of a highly successful standard such as **BS 1363 for electrical plugs and sockets**.

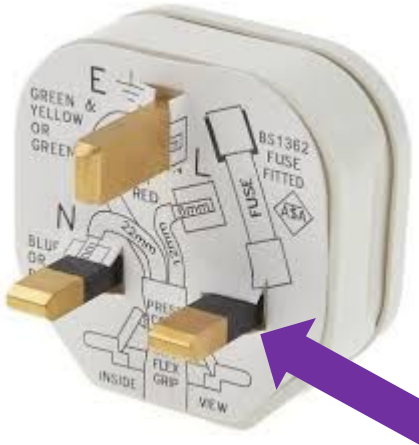
You could mention ***compulsory features*** such as ***strain relief, fuses and earth-pin operated shutters***.

BS 1363 for electrical plugs and sockets



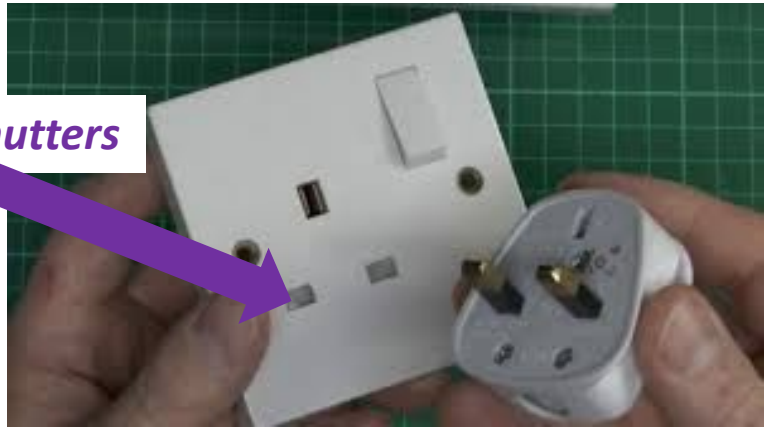
Strain Relief

Fuses



Live and Neutral Pin Insulation

Earth-pin Operated Shutters



Now Test Yourself

1. *Which organisation uses a kitemark to indicate products that meet appropriate safety and quality standards?*
2. *What mark would you expect to see on a product that meets European Standards?*
3. *Which British Standards document would a UK 3-pin 13 amp plug manufacturer refer to and comply with in order to reach the required standards of safety?*
4. *Explain briefly why it is necessary to have national and international standards for products and services.*